Contextual Attention Transduction for Question Answering and Semantic Extraction

Dr. Sarah Lin and Dr. David Chen, Member, IEEE

Abstract- *Bidirectional encoder representations offer rich sentence embeddings for reading comprehension.*

I. INTRODUCTION

Modern computer vision models have experienced dramatic evolution. Developing architectures that balance latency and feature selectivity remains a central topic of inquiry in machine intelligence.

II. RELATED WORK

Prior approaches often struggled with dense multi-scale object distributions.

III. PROPOSED METHODOLOGY

In this section, we delineate our proposed framework. The core system architecture employs BERT Transformer with Rotary Positional Embeddings. We integrate asymptotic multi-level feature pyramid routing, connecting cross-layer feature maps with learned convolutional kernels. Training is executed over the SQuAD 2.0 benchmark. The loss formulation balances bounding-box focal regression with cross-entropy classification.

IV. EXPERIMENTAL RESULTS

To evaluate the proposed model, extensive experiments were conducted on SQuAD 2.0. The proposed architecture achieved 88.7% F1 accuracy. The model achieves high linguistic understanding across dense textual passages.

V. CONCLUSION

In this work, we introduced a novel framework that improves upon established baselines.